ONLINE RETAIL SALES ANALYTICS REPORT

Turning Retail Data Into Actionable Business Insight

SQL-Powered Retail Analytics Project

Prepared For

Retail Business Analytics Client

Prepared By

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Tools & Technologies

MySQL | SQL | Database Design | Business Analytics

2026

Executive Summary

Project Overview

This report analyzes an online retail transaction dataset using SQL to extract meaningful business insights. The analysis focuses on sales performance, customer behavior, product trends, and market performance.

Key Analysis Performed

✓ MySQL database design and organization
✓ Sales and revenue analysis
✓ Customer behavior analysis
✓ Product performance evaluation
✓ Country-wise sales analysis
✓ Business KPI analysis

Business Value Delivered

The analysis helps identify revenue opportunities, understand customer patterns, optimize products, and support data-driven business decisions.

Dataset Overview

Dataset Description

This project uses an online retail transaction dataset containing customer purchases, product details, order information, and

revenue data. The dataset was analyzed to identify sales trends, customer behavior, product performance, and market insights.

Dataset Information

Dataset Type: Transactional Retail Sales Data
Database Platform: MySQL

Total Transactions: 406,431
Unique Customers: 4,372
Total Orders: 22,190
Total Revenue: 8,401,284.13

Key Data Attributes

Column	Description
InvoiceNo	Unique identification number for each order
InvoiceDate	Date and time of transaction
CustomerID	Unique customer identifier
Country	Customer location
StockCode	Product identification code
Description	Product name/details
Quantity	Number of items purchased
UnitPrice	Price per unit
Revenue	Calculated transaction revenue

Data Preparation

The dataset was cleaned and structured to ensure accuracy and consistency. Key steps included handling missing values, standardizing date formats, and preparing the database for SQL analysis.

4. Database Design & Architecture

4.1 Database Structure Overview

A relational database was designed using MySQL to organize customer, product, and transaction data efficiently. The structure reduces redundancy and enables effective SQL-based analysis.

The database consists of four main tables:

- **Customers**
- **Products**
- **Orders**
- **Order Details**

4.2 Database Schema Design

Customers Table

Stores customer information.

Primary Key: CustomerI

The Customers table stores customer-related information.**Primary Key:** CustomerID

Field	Description
CustomerID	Unique identifier for each customer
Country	Customer location

Products Table

The Products table contains product information.

Primary Key: StockCode

Field	Description
StockCode	Unique product identifier
Description	Product name
UnitPrice	Price per product unit

Orders Table

The Orders table stores order-level transaction information.

Primary Key: InvoiceNo

FIELD	DESCRIPTION
INVOICENO	Unique order identifier
INVOICEDATE	Date and time of purchase
CUSTOMERID	Customer reference

Order Details Table

The Order Details table contains item-level purchase information.

Foreign Keys: InvoiceNo, StockCode

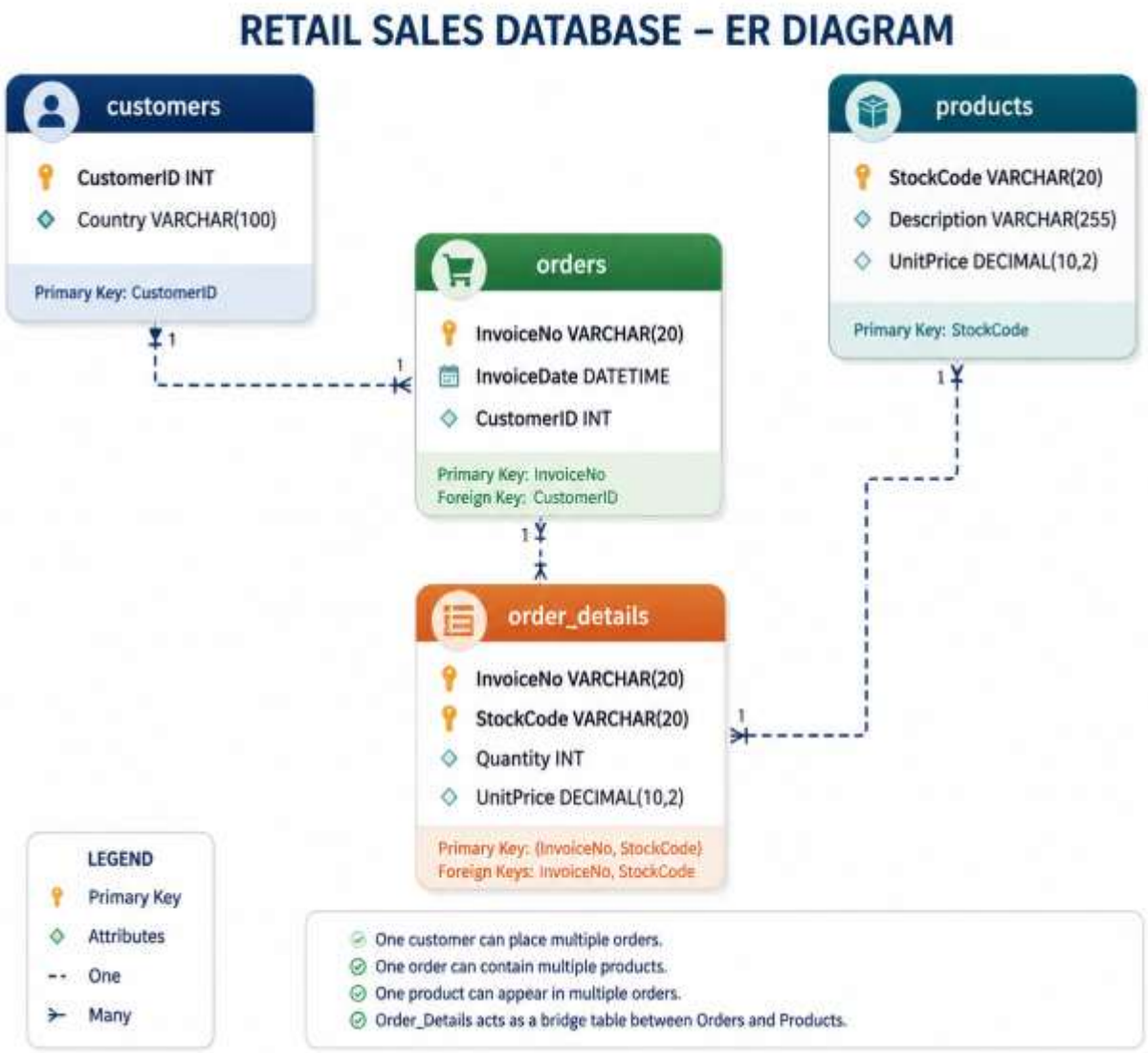
Field	Description
InvoiceNo	Related order number
StockCode	Related product code
Quantity	Number of items purchased
UnitPrice	Price per unit

4.3 Entity Relationship Diagram (ERD)

The database relationship structure is represented through an Entity Relationship Diagram (ERD).

The relationships include:

- One customer can place multiple orders.
- One order can contain multiple products.
- One product can appear in multiple orders.
- Order Details acts as a bridge table between orders and products.



4.4 Database Design Benefits

Database Design Benefits

The relational database design provides:

✓ Better data organization
✓ Reduced data redundancy
✓ Efficient SQL analysis
✓ Easy maintenance
✓ Future scalability

5. Data Cleaning & Preparation Process

5.1 Data Preparation Overview

The raw dataset was cleaned and transformed into an analysis-ready format. Data preparation included handling missing values, standardizing formats, and creating required calculated fields.

5.2 Data Cleaning Steps

Missing Values Handling

Identified and managed missing records to improve data quality.

Date Standardization
Converted transaction dates into a consistent format for time-based analysis.

Data Consistency Check
Reviewed data for inconsistencies and duplicate records.

Revenue Calculation

Revenue = Quantity × UnitPrice

Used for sales, customer, and product performance analysis.

5.3 Final Analysis Dataset Summary

After cleaning and transformation, the dataset was prepared for SQL-based business analysis, enabling accurate insights into sales, customers, products, and market performance.

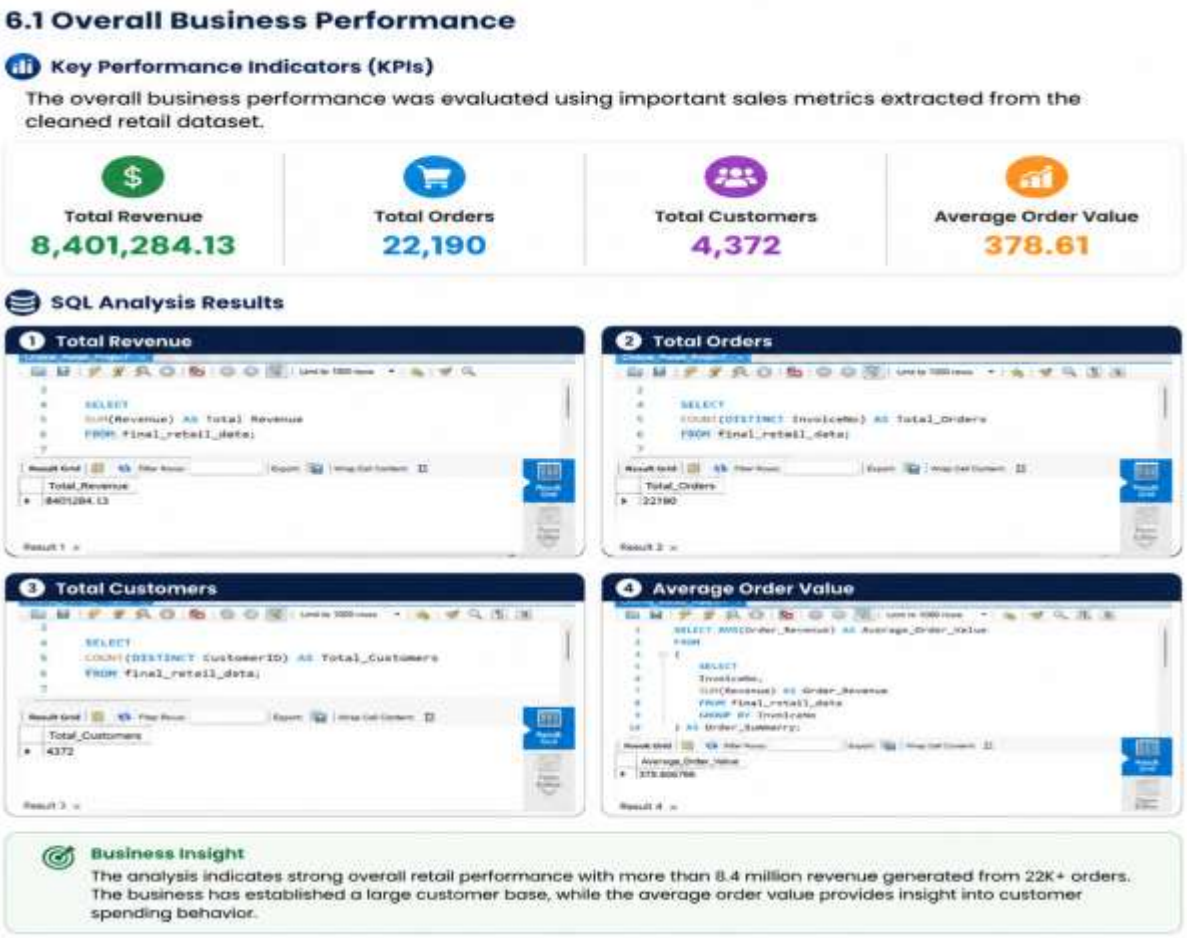
Final Dataset Statistics

METRIC	VALUE
TOTAL TRANSACTION RECORDS	406,431
UNIQUE CUSTOMERS	4,372
TOTAL ORDERS	22,190

TOTAL REVENUE	8,401,284.13
DATABASE PLATFORM	MySQL

The prepared dataset enabled efficient analysis and helped identify key business opportunities through structured SQL-based reporting.

Step 6: Business Analysis & Key Insights



Business Insight

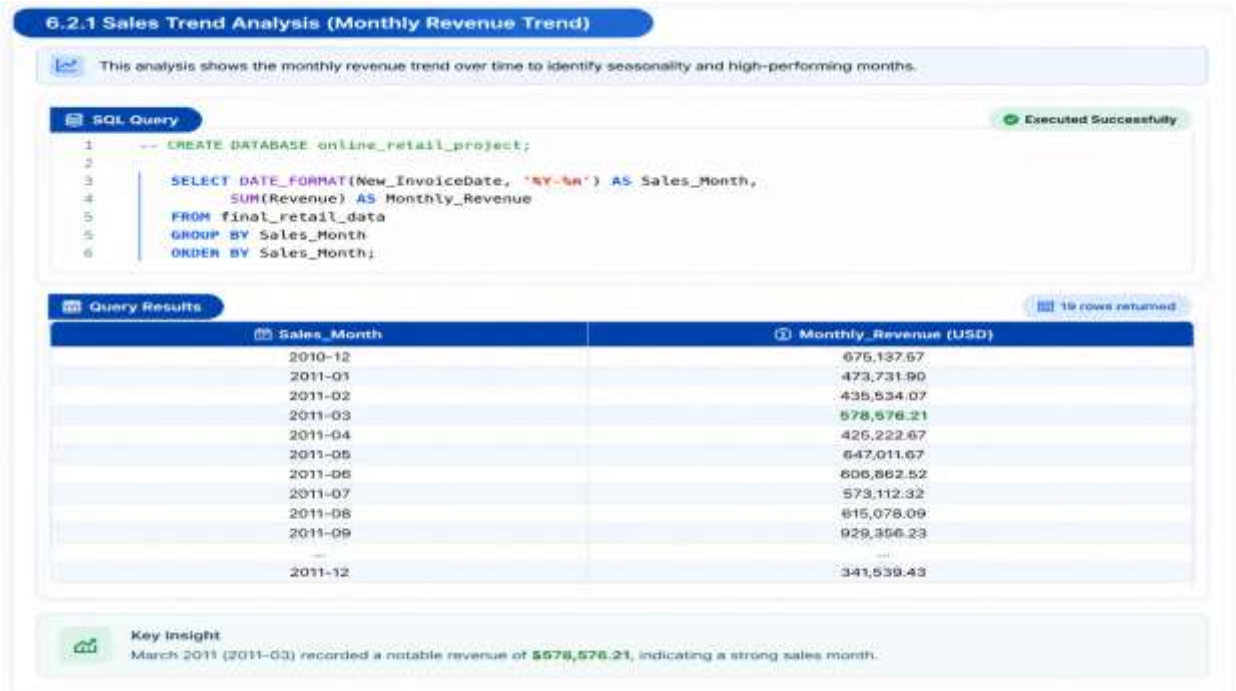
The analysis indicates strong overall retail performance with more than **8.4 million revenue** generated from **22K+ orders**. The business has established a large customer base, while the average order value provides insight into customer spending behavior.

6.2.1 Sales Trend Analysis

Monthly revenue analysis highlights sales patterns and seasonal variations over time.

Key Insights:

- Identifies peak sales periods.
- Supports sales forecasting and inventory planning

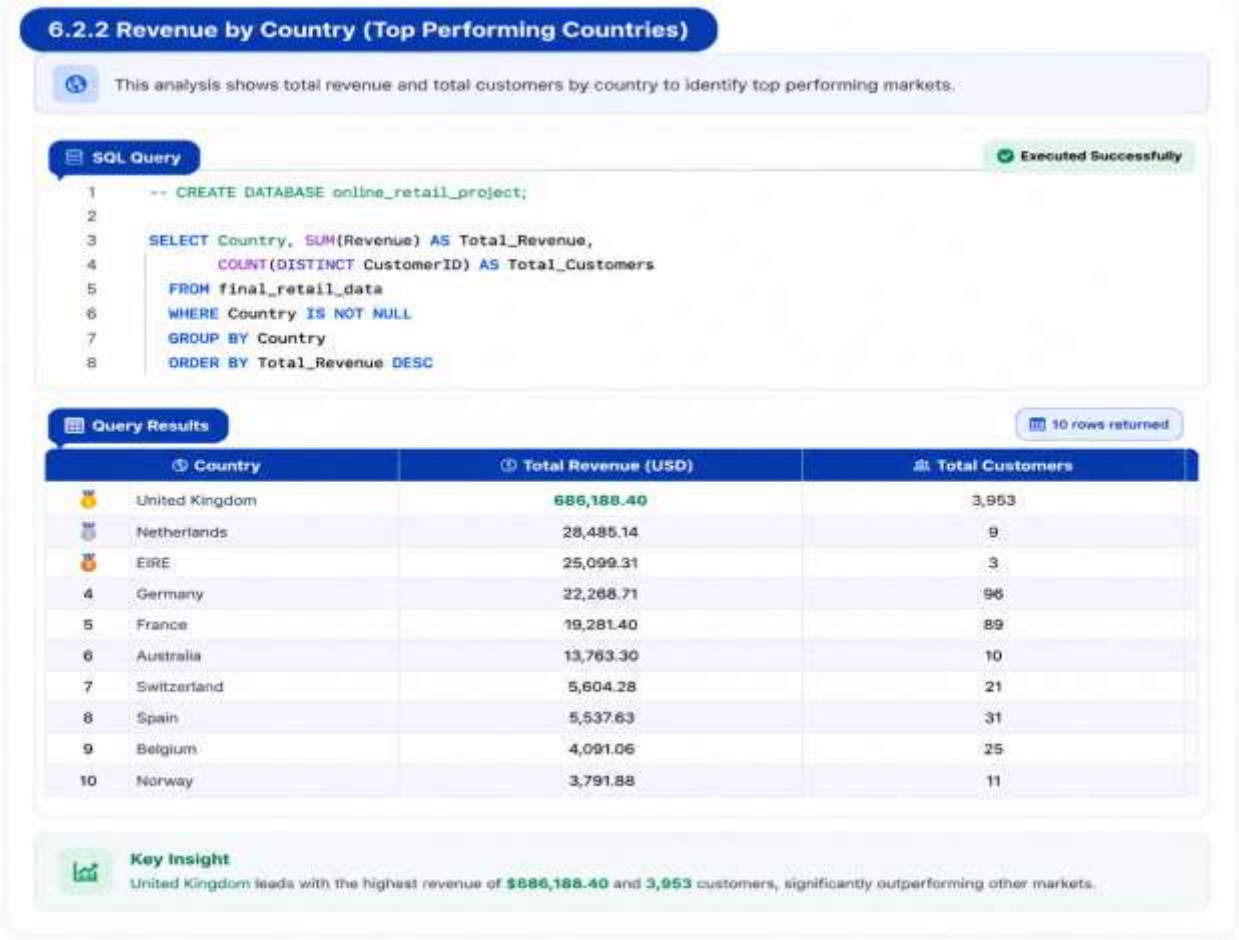


6.2.2 Geographic Sales Performance

Country-wise analysis identifies the most valuable markets. The **United Kingdom generated the highest revenue**, making it the primary market for the business.

Key Insights:

- UK is the leading revenue contributor.
- Market analysis supports regional growth strategies.



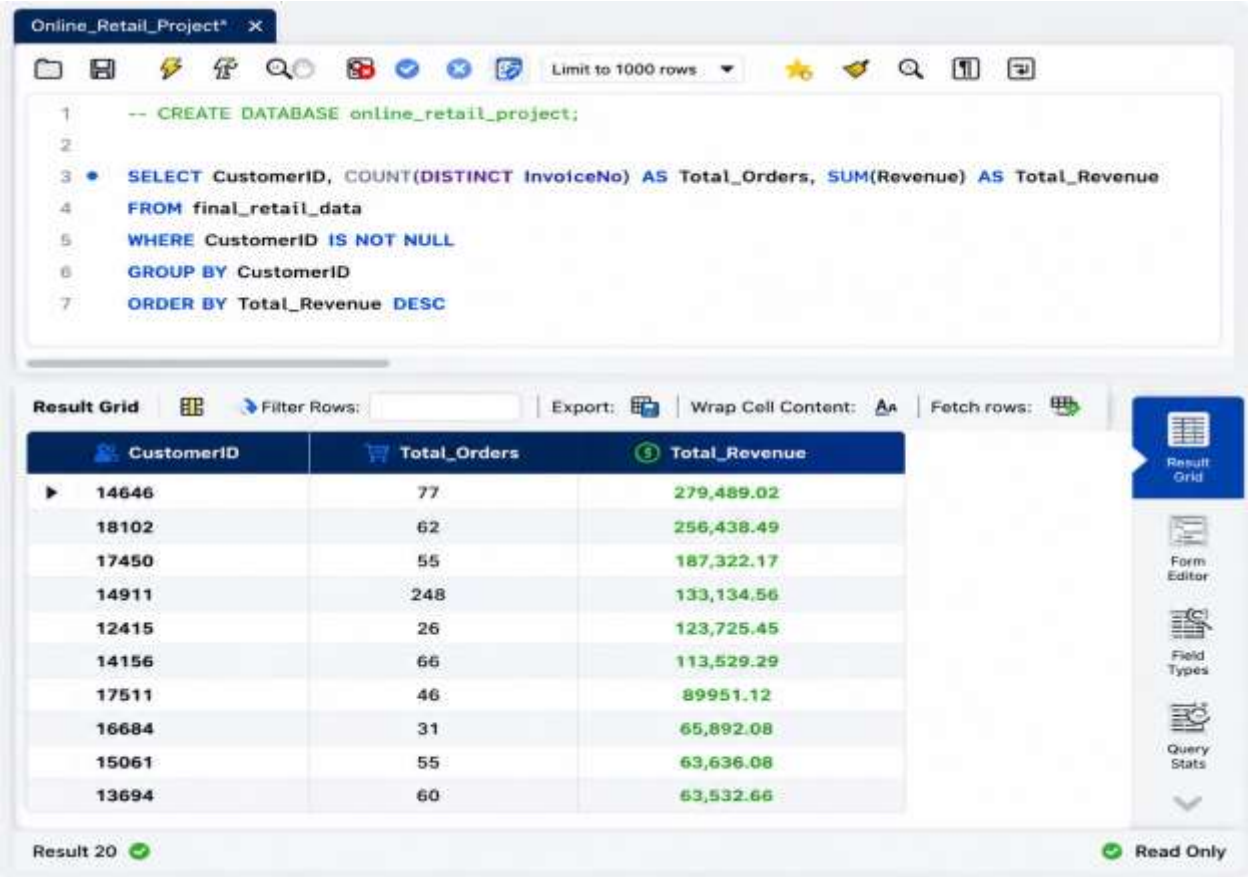
6.3 Customer Analysis

6.3.1 Top Customer Analysis

Customer-level analysis was performed to identify the highest revenue-generating customers. The results highlight valuable customers who contribute significantly to overall business revenue.

Key Insights:

- Identifies high-value customers.
- Supports customer retention and personalized marketing strategies.
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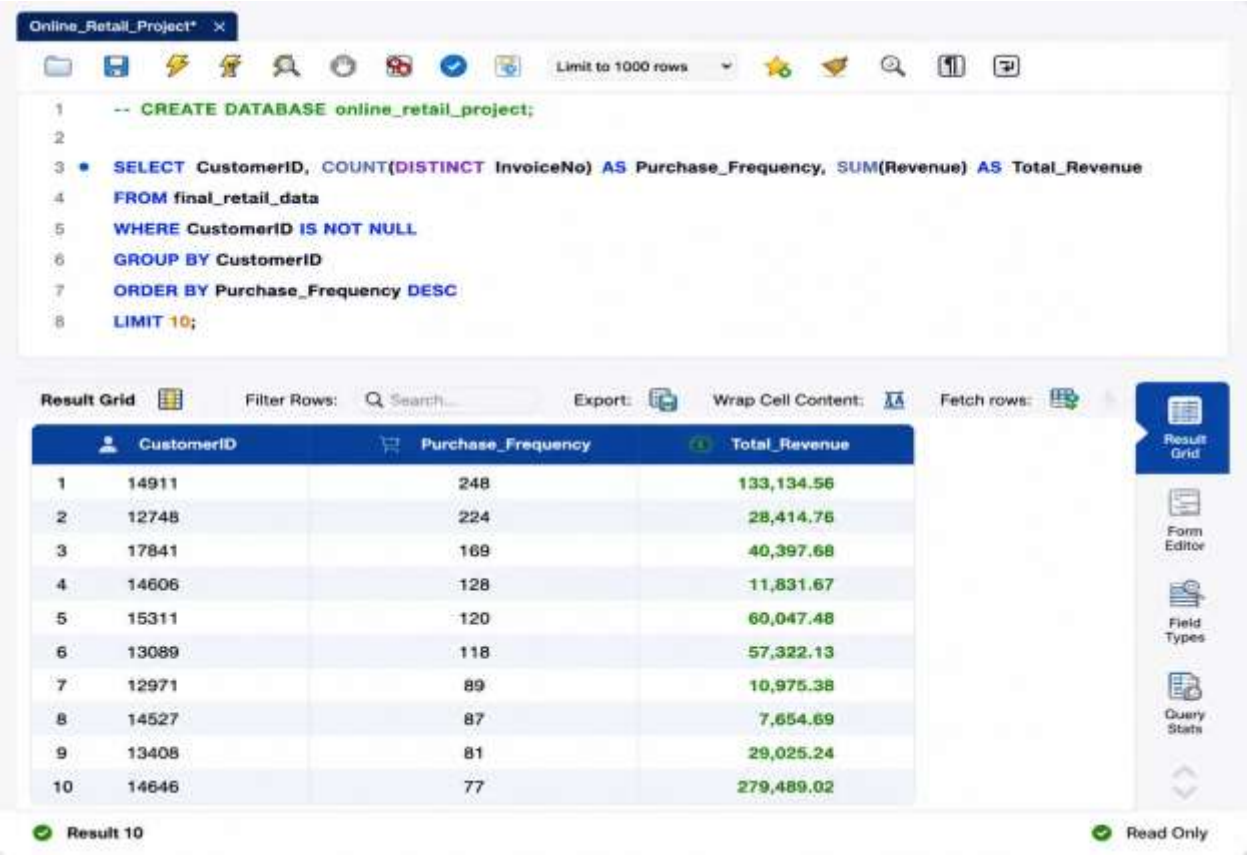
6.3.2 Customer Purchase Frequency Analysis

Customer Purchase Frequency Analysis

Purchase frequency analysis evaluates customer ordering behavior and identifies repeat buyers. Frequent customers represent strong engagement and potential long-term value.

Key Insights:

- Helps identify loyal customers.
- Supports retention strategies and targeted promotions.



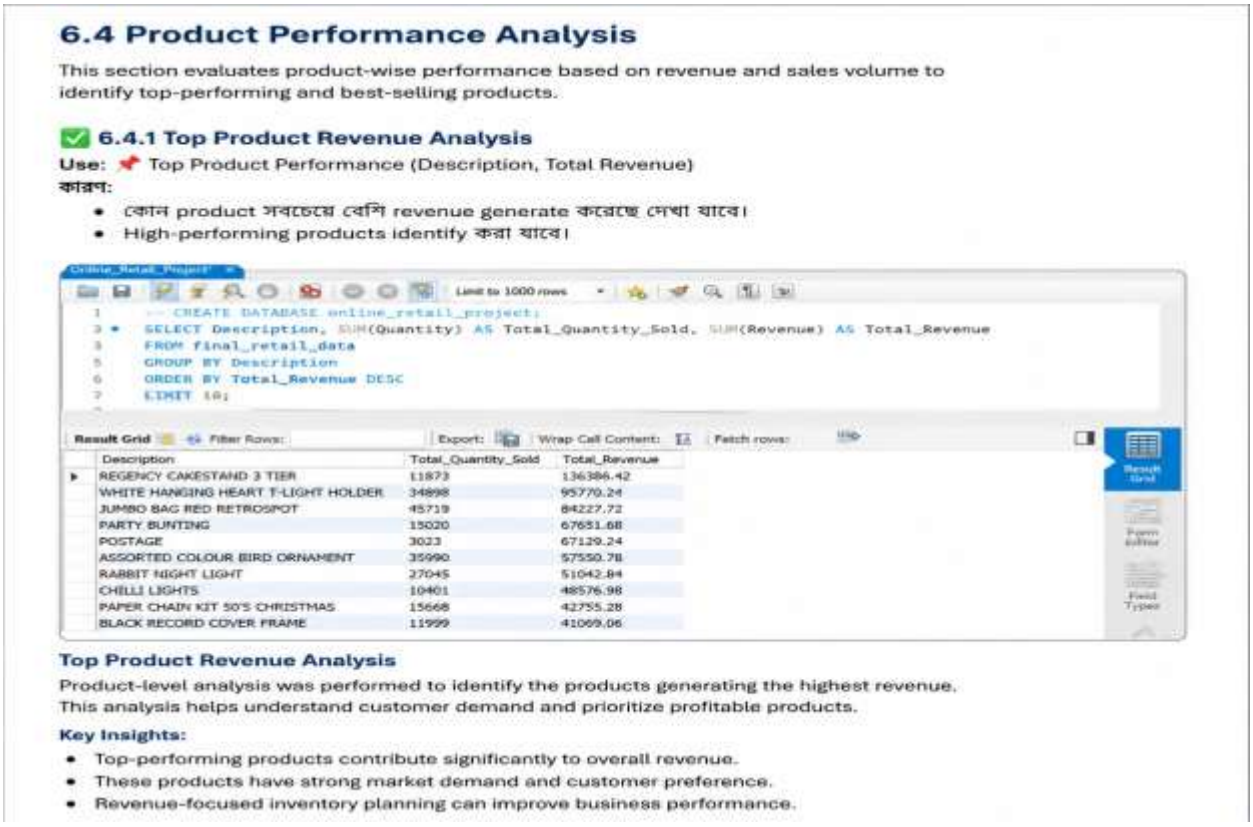
6.4 Product Performance Analysis

6.4.1 Top Product Revenue Analysis

Product-level analysis was performed to identify the products generating the highest revenue. This analysis helps understand customer demand and prioritize profitable products.

Key Insights:

- Top-performing products contribute significantly to overall revenue.
- Revenue-focused inventory planning can improve business performance.



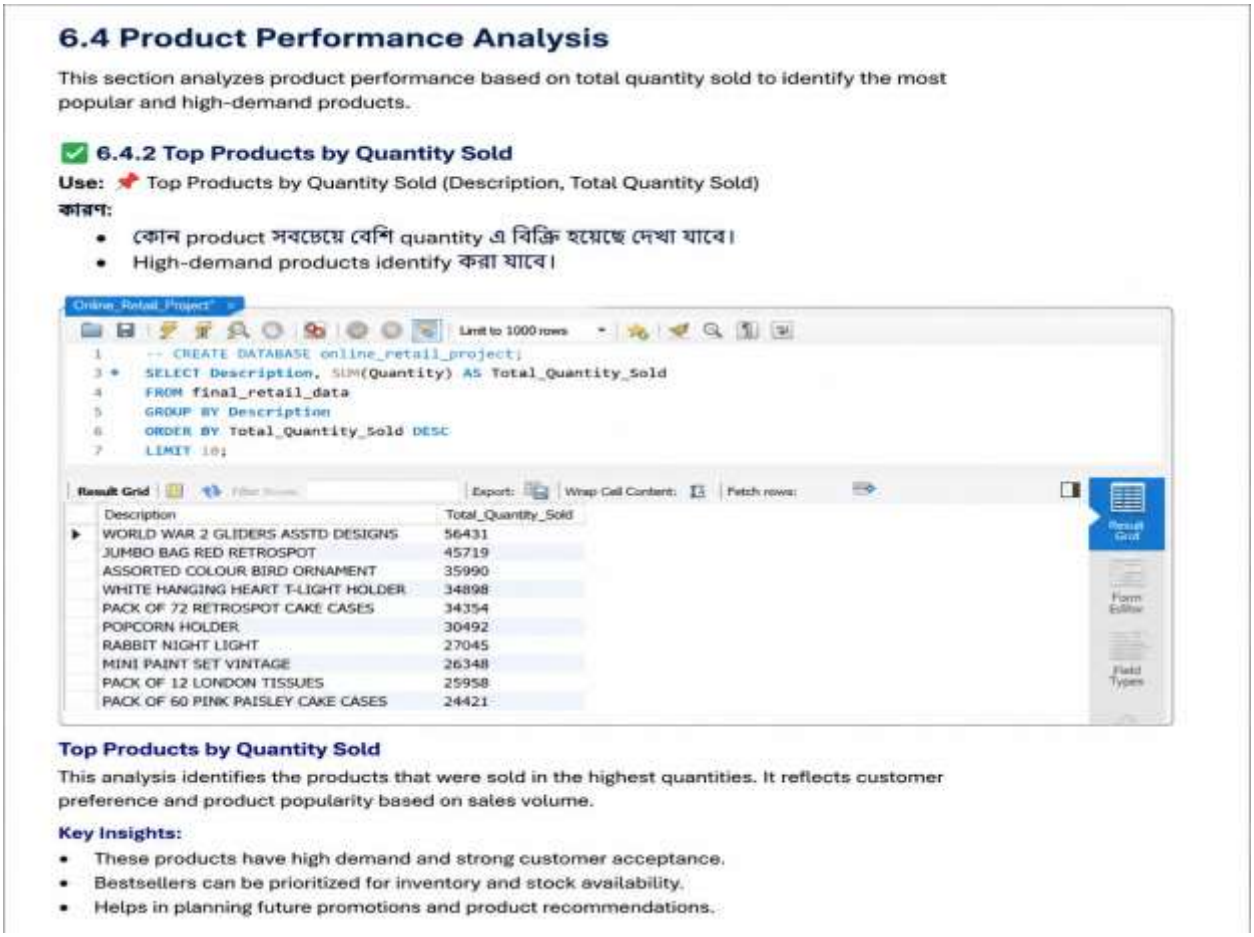
6.4.2 Best Selling Product Analysis

Best Selling Product Analysis

The best-selling product analysis identifies products with the highest sales volume based on total quantity sold. This helps understand customer preferences and product demand.

Key Insights:

- High-demand products should maintain sufficient stock availability.
- Sales volume analysis supports better inventory decisions.



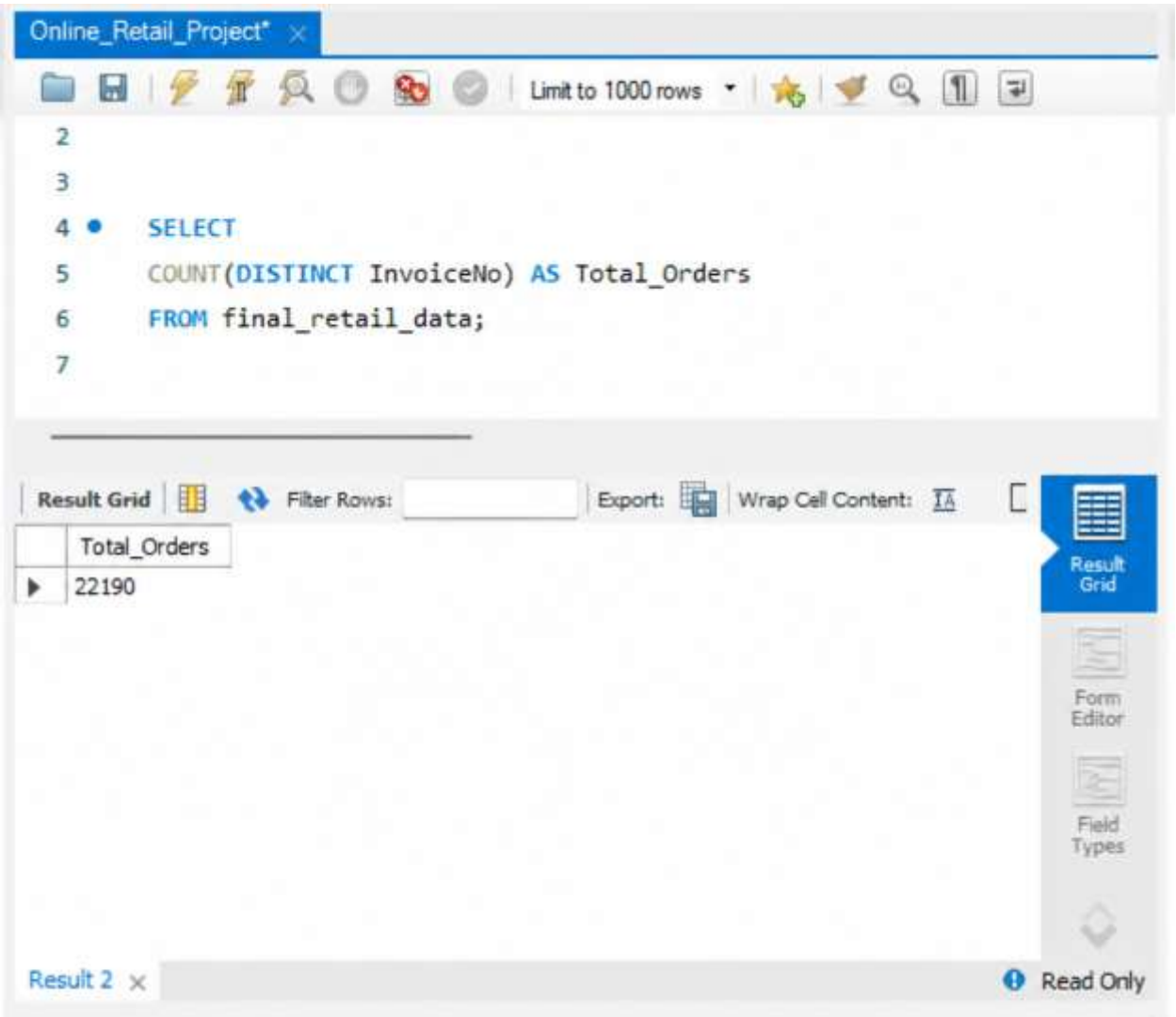
6.6 Order Analysis and Cancellation Analysis

6.6.1 Overall Order Performance

Order analysis evaluates the overall transaction activity of the retail business.

Key Insights:

- Total orders: 22,190
- Indicates strong customer purchasing activity.

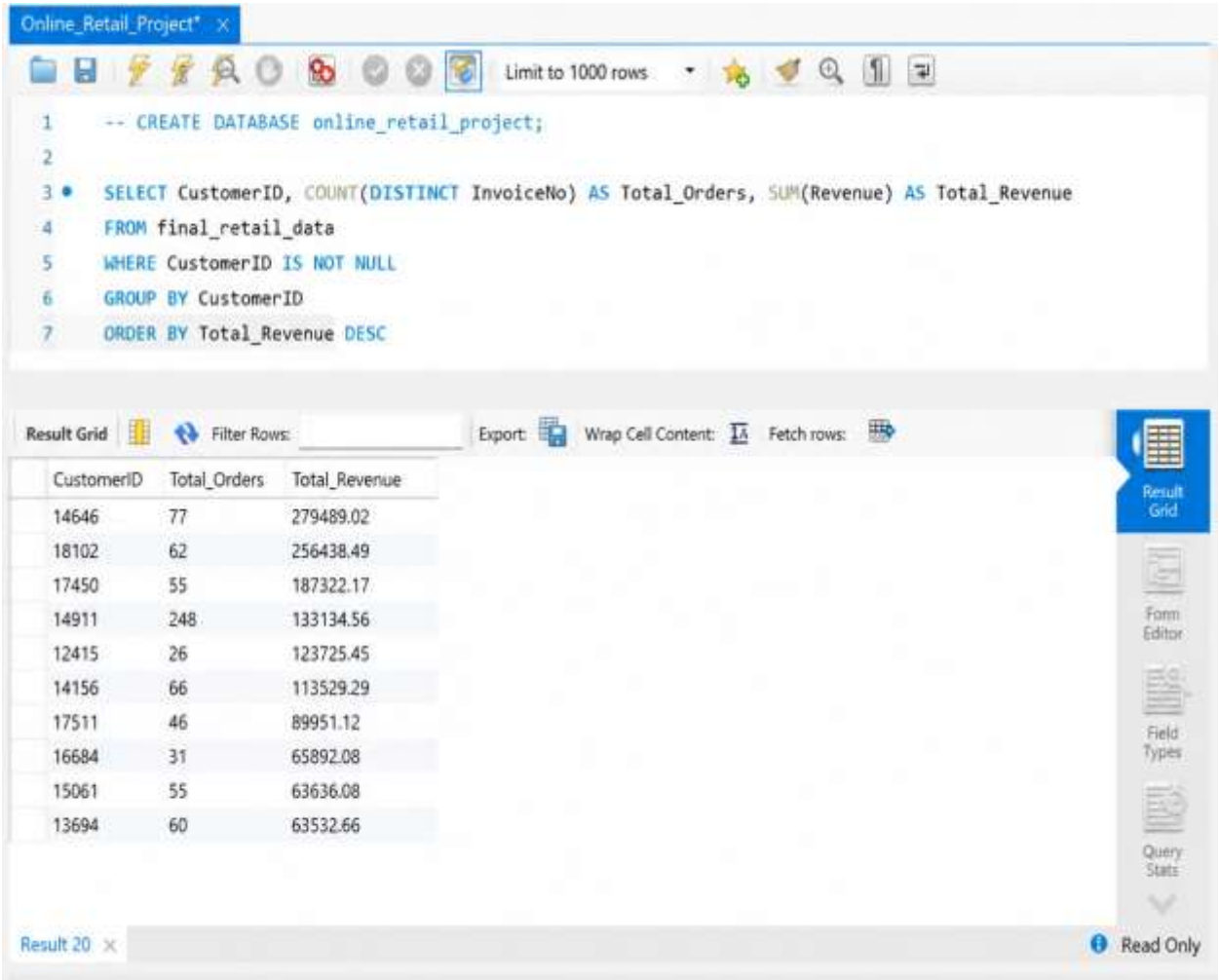


6.6.2 Top Customer Order Contribution

Customer analysis identifies high-value customers based on order and revenue contribution.

Key Insights:

- Top customers generate significant revenue.
- Helps improve customer retention strategies..

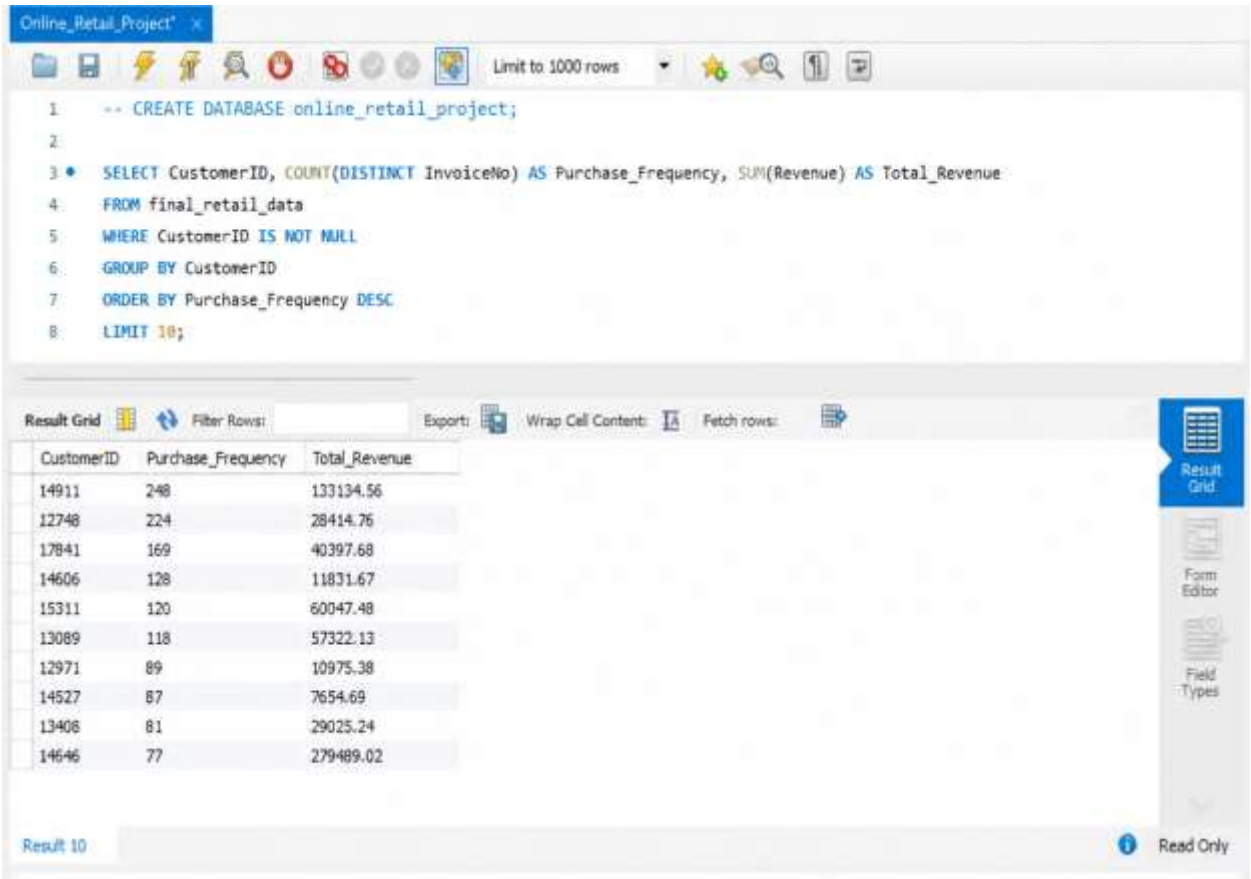


6.6.3 Customer Purchase Frequency Analysis

Purchase frequency analysis measures repeat buying behavior.

Key Insights:

- Frequent buyers represent loyal customers.
- Supports targeted marketing campaigns.

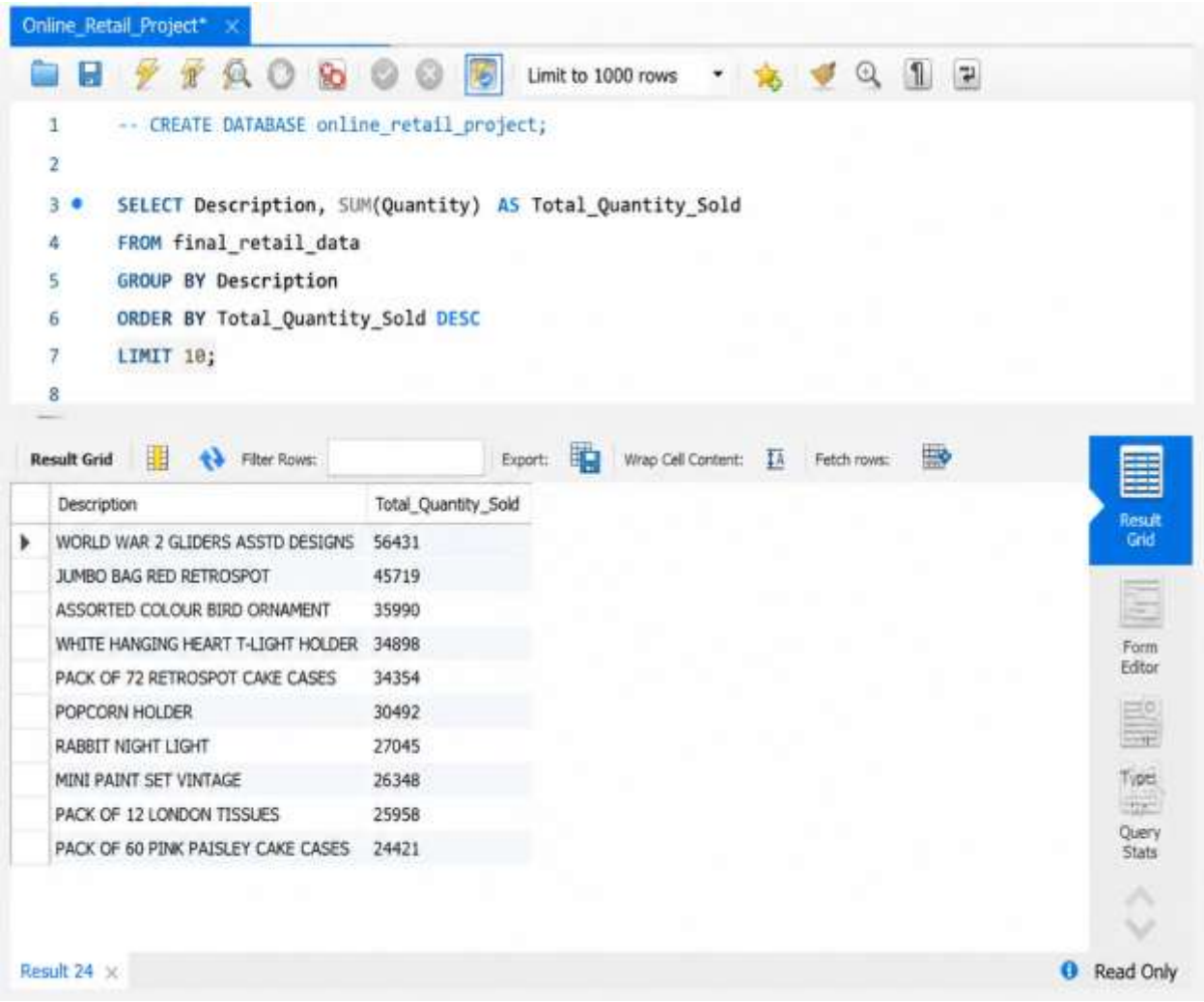


6.6.4 Best-Selling Product Analysis

Product order analysis identifies items with the highest customer demand.

Key Insights:

- Best-selling products drive sales performance.
- Helps optimize inventory decisions.



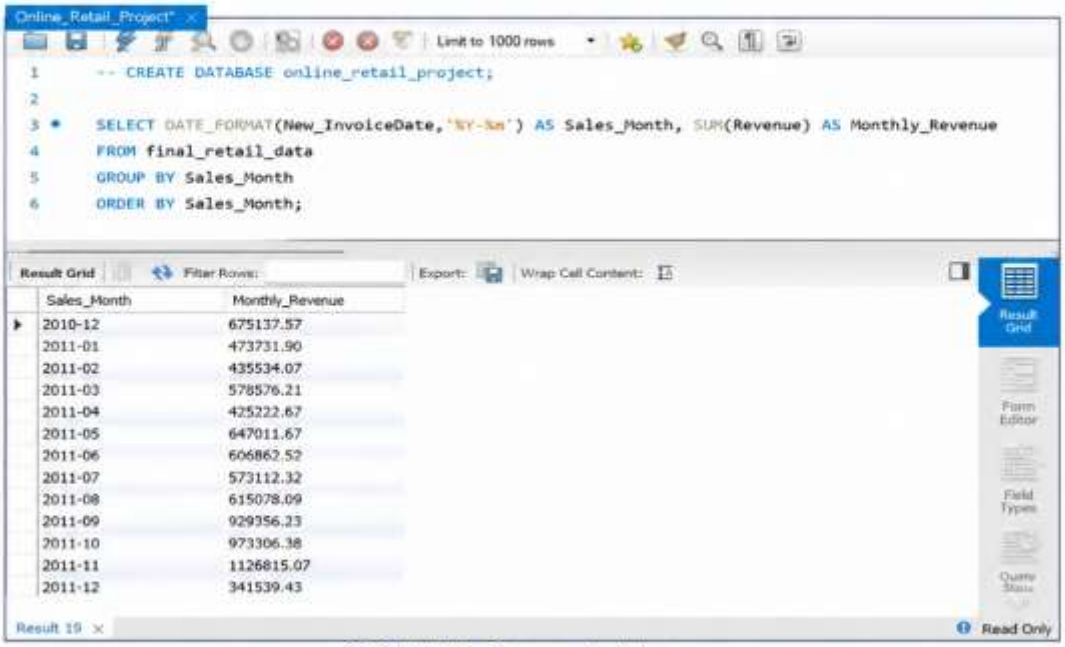
6.7 Sales Trend Analysis

6.7.1 Monthly Sales Trend Analysis

Monthly sales trend analysis evaluates revenue variations across different months and identifies peak and low sales periods.

Key Insights:

- Revenue fluctuated across months.
- Highest revenue was recorded in October and November 2011.
- Helps in planning inventory and marketing strategies.



```
-- CREATE DATABASE online_retail_project;

SELECT DATE_FORMAT(New_InvoiceDate,'%Y-%m') AS Sales_Month, SUM(Revenue) AS Monthly_Revenue
FROM final_retail_data
GROUP BY Sales_Month
ORDER BY Sales_Month;
```

Sales_Month	Monthly_Revenue
2010-12	675137.57
2011-01	473731.90
2011-02	435534.07
2011-03	578576.21
2011-04	425222.67
2011-05	647011.67
2011-06	606862.52
2011-07	573112.32
2011-08	615078.09
2011-09	929356.23
2011-10	973306.38
2011-11	1126815.07
2011-12	341539.43

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6.7.2 Overall Sales Performance Summary

Key Insights:

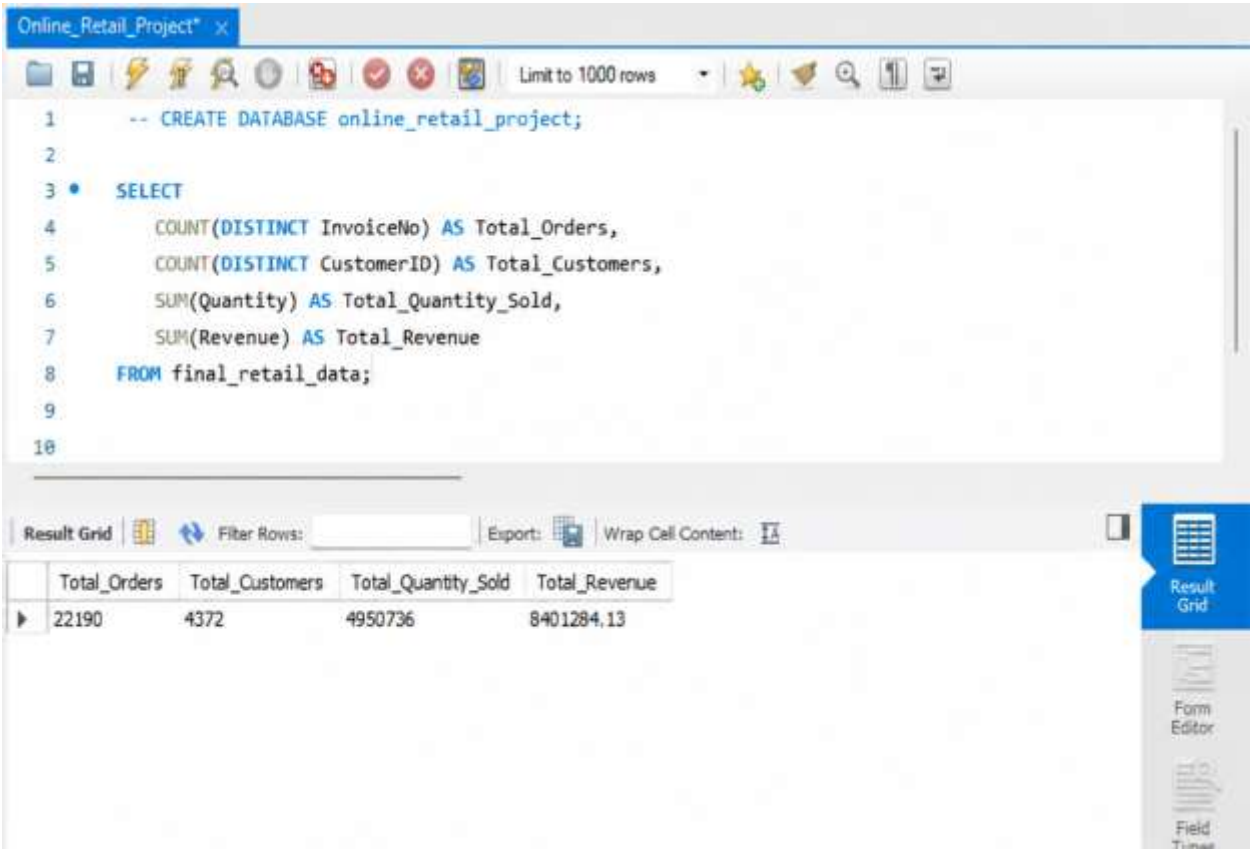
- Total Revenue Generated: **8,401,284.13**
- Total Number of Orders: **22,190**
- Customer and product performance drive overall business growth.

6.7.2 Sales Performance Overview

Sales performance analysis summarizes overall business performance by evaluating revenue trends and customer demand patterns.

Key Insights:

- Provides an overview of business growth.
- Helps identify improvement opportunities.
- Supports future decision-making.

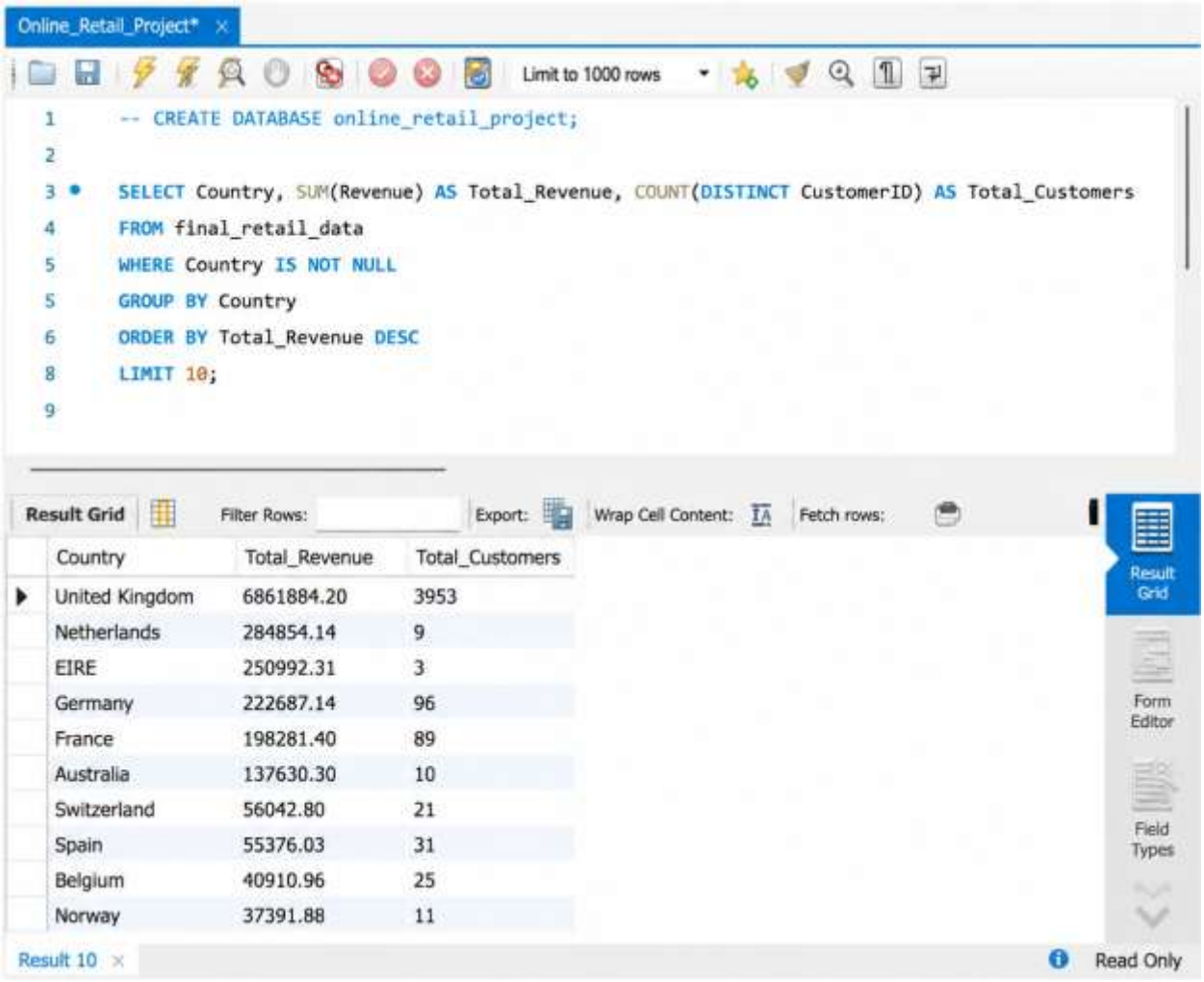


6.7.3 Country-wise Sales Performance Analysis

Country-wise sales analysis evaluates revenue contribution and customer distribution across different countries.

Key Insights:

- The United Kingdom generated the highest revenue (6,861,884.20) with the largest customer base.
- Other countries contributed comparatively lower revenue.
- Helps identify major markets and supports regional business strategies.

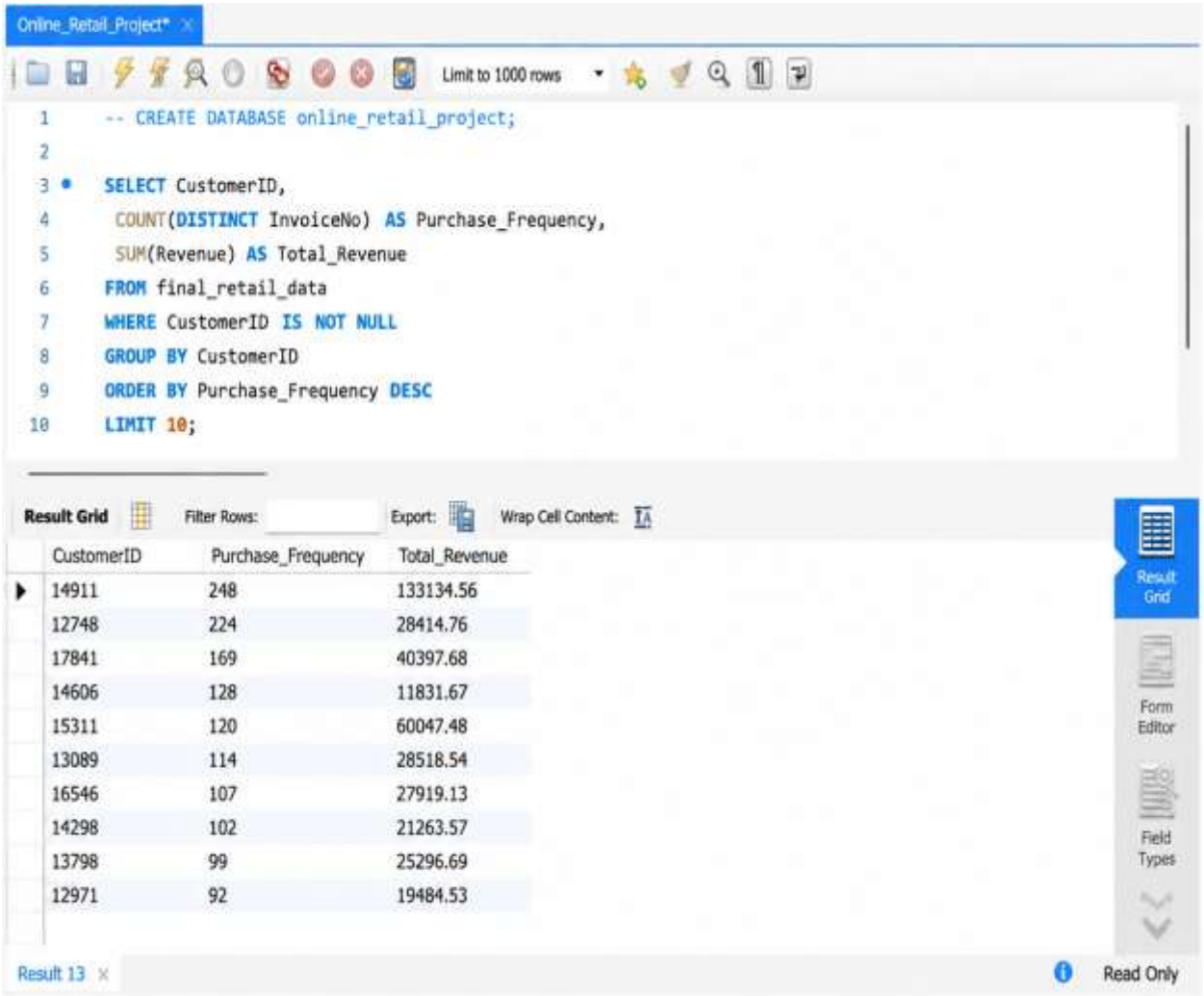


6.7.4 Customer Purchase Behavior Analysis

Customer purchase analysis evaluates buying patterns by measuring order frequency and customer revenue contribution.

Key Insights:

- Identifies high-value and frequent customers.
- Helps understand customer purchasing trends.
- Supports customer retention and marketing strategies.
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7. Product Performance Analysis

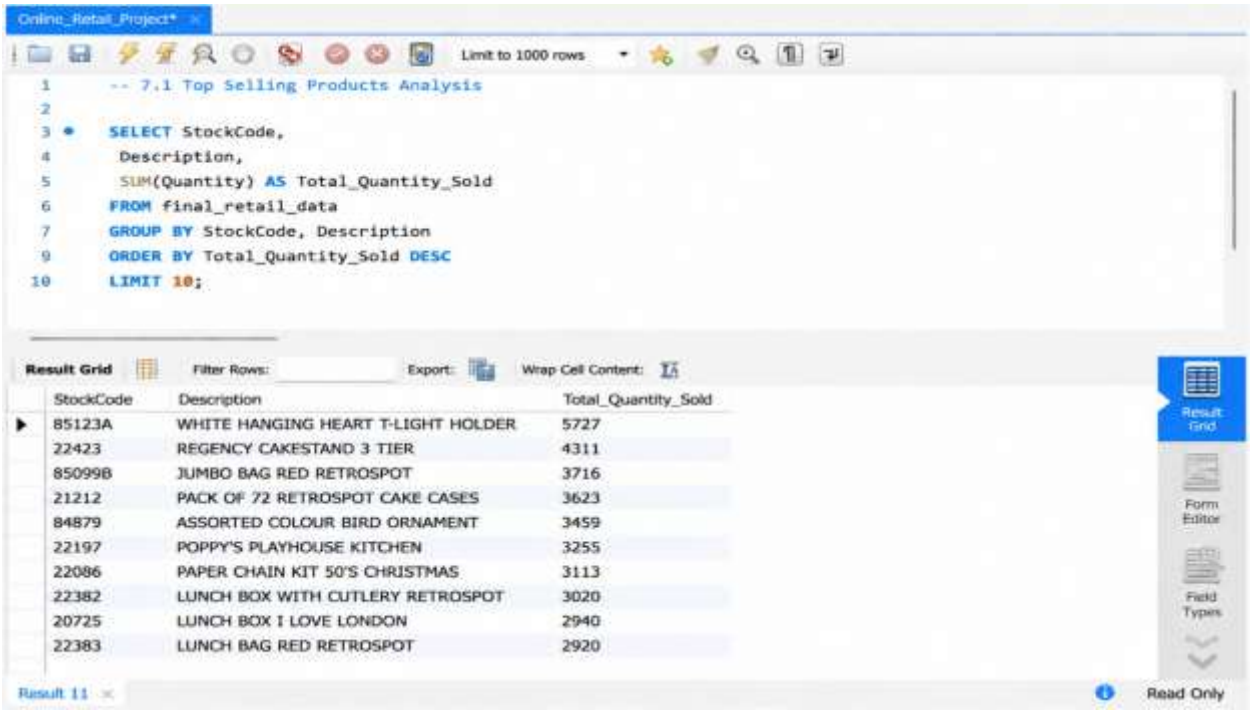
Product analysis identifies high-performing products based on sales volume and revenue contribution.

7.1 Top Selling Products Analysis

Identifies products with the highest sales quantity.

Key Insights:

- Shows customer demand patterns.
- Helps improve inventory planning.

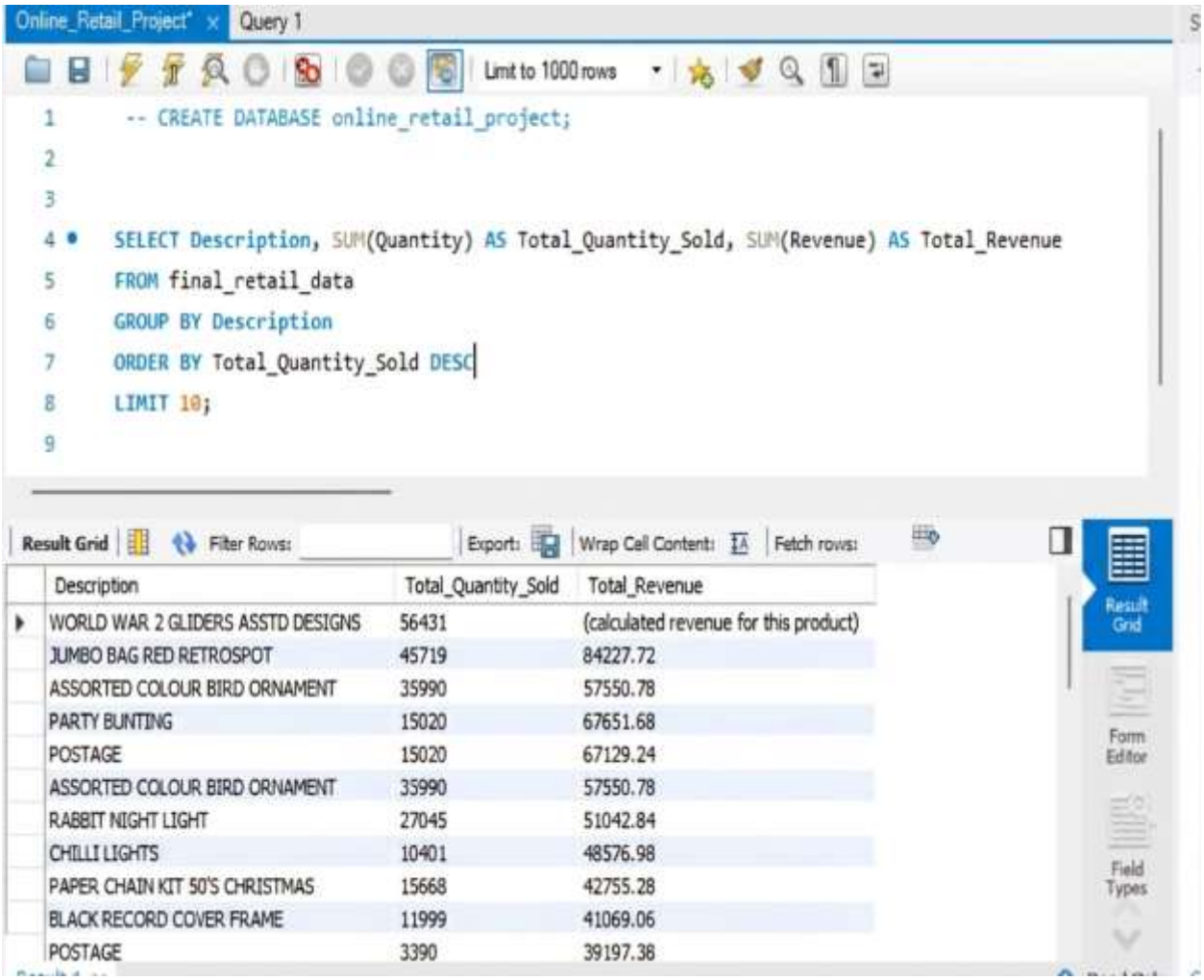


7.2 Product Revenue Analysis

Analyzes products generating the highest revenue.

Key Insights:

- Identifies profitable products.
- Supports sales and marketing decisions.

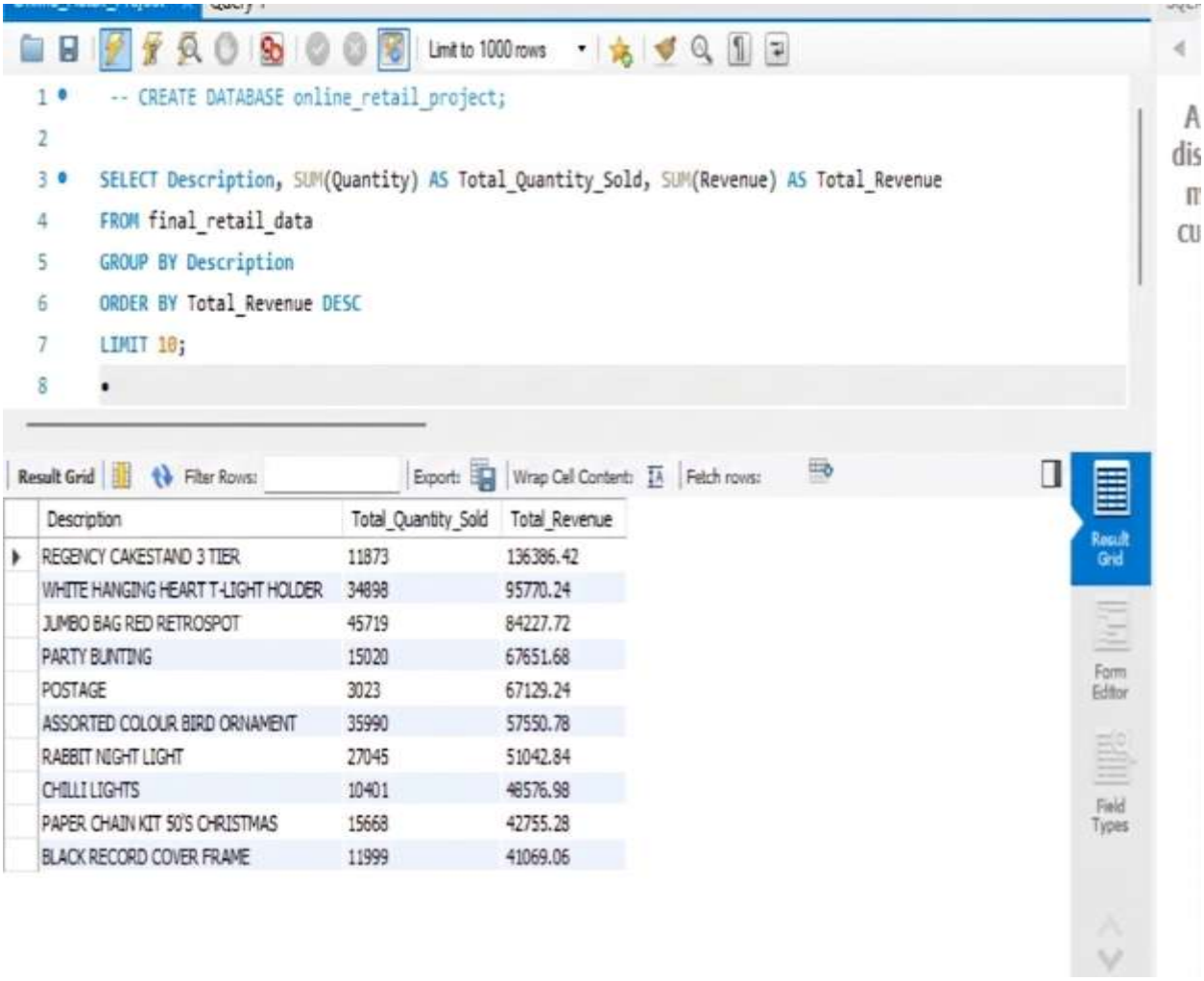


7.3 Product Performance Insights

Combines sales volume and revenue data to evaluate product effectiveness.

Key Insights:

- Prioritizes successful products.
- Helps optimize future business strategies.



8. Business Insights & Recommendations

This section summarizes the key findings from the sales, customer, and market analysis and provides recommendations for improving business performance.

8.1 Key Business Insights

The analysis highlights important patterns in sales performance, customer behavior, and revenue contribution.

Key Insights:

- Overall sales performance shows strong revenue generation with consistent customer demand.
- A small number of customers contribute significantly to total revenue.
- High-performing products play an important role in driving sales.
- The United Kingdom represents the largest revenue-generating market.

8.2 Business Recommendations

Based on the analysis findings, the following recommendations can improve business efficiency and growth.

Recommendations:

- Maintain sufficient inventory for high-demand products.

- Focus marketing efforts on valuable and frequent customers.
- Develop targeted strategies for different markets.
- Promote profitable products to increase revenue.

8.3 Conclusion

The analysis provides valuable insights into sales trends, customer behavior, and business performance. These findings can support data-driven decision-making and improve future business strategies.

9. Conclusion

This project successfully transformed raw online retail transaction data into meaningful business insights using SQL-based analysis.

The analysis covered sales performance, customer behavior, product trends, and market performance to identify key business opportunities.

The findings demonstrate how data-driven analysis can support better decision-making, improve operational strategies, and enhance overall business performance.